

7.7.6.3 PBR Message Routing within a Single PBR Switch

A message received or converted to PBR format at a PBR switch ingress port is routed to one of the switch's egress ports, as determined by the ingress port's DPID Routing Table (DRT) and its associated Routing Group Table (RGT). Their structures are described in detail in [Section 7.7.13.10](#) and [Section 7.7.13.12](#), respectively, and this section provides a high-level summary.

A DRT has 4096 entries and is indexed by a DPID. Each DRT entry contains a 2-bit entry type field that indicates whether the entry is valid, and whether the entry contains a single physical port number or an RGT index.

DRT entries that contain an RGT index are required when multiple egress ports need to be specified for use with dynamic routing. An RGT is a power-of-2-sized table with up to 256 entries. Each RGT entry contains an ordered list of up to eight physical port numbers, along with two 3-bit fields that indicate how many in the list are valid and how many of those are primary vs. secondary. This allows one or more primary and zero or more secondary egress ports to be listed. Cases that require static routing must always use the first list entry. The RGT entry also contains a 3-bit dynamic routing mode and 3-bit mix setting. The distinction between primary vs. secondary varies by dynamic routing mode and mix setting.

In routing modes that utilize the mix setting, its value determines the mix of the primary and secondary egress port group usage, assuming that one or more secondary egress ports are specified. Using the mix setting supports egress port selection based on known bandwidth differences that exist elsewhere in the fabric or based on preferred vs. overflow routing paths. Secondary egress ports should be specified only when there are significant differences with primary egress ports; otherwise, all suitable egress ports should be specified as primary. When no secondary egress ports have been specified, the mix setting shall be ignored.

Mix Setting	% Primary	% Secondary
0	87.5	12.5
1	75	25
2	62.5	37.5
3	50	50
4	37.5	62.5
5	25	75
6	12.5	87.5
7	Preferred	Overflow

Mix setting 7 is intended for use in cases where primary and secondary egress port groups represent preferred and overflow ports, respectively. Mix setting 7 mandates the choice of a primary (preferred) path route whenever flow-control conditions and link state permit.

The term candidate egress port refers to a port that is present in the appropriate RGT entry, where the message can be queued or internally routed immediately. The egress port need not have link credits to send the packet immediately. An implementation may optionally base part of the candidate selection on the egress port state (e.g., link-up or containment states).

The mix dynamic routing mode descriptions that follow describe routing outcomes in terms of probability, consistent with a weighted (pseudo) random implementation. Random selection has the advantage that each routing decision is stateless and independent of one another, and it has high immunity to hot-route problems that might

otherwise arise from repetitive patterns in packet arrivals. The specific random routing implementation is not prescribed. Implementations that achieve the specified mix by deterministic means, such as by weighted round-robin, are permitted.

The architected dynamic routing modes include the optional modes listed in [Table 7-85](#).

Table 7-85. Optional Architected Dynamic Routing Modes

Mode	Description
Mix with Random	The candidate list is first narrowed to select either the primary or the secondary group based on the configured mix. A random selection is then made within that group. A message class shall stall when the selected subset is empty due to flow-control conditions. The FM may choose to select this mode (if supported) as an alternative to Mix with Congestion Avoidance if the latter is not supported.
Mix with Congestion Avoidance	The candidate list is first narrowed to select either the primary or the secondary group based on the configured mix. A local congestion-avoiding selection is then made within that group. A message class shall stall when the selected subset is empty due to flow-control conditions. Congestion-avoiding candidate selection is based on vendor-specific congestion metrics, favoring the selection of less-congested egress ports. For example, the congestion metric might be a measure of egress port backlog, considering all queued traffic for that egress port across the entire switch. The FM may choose to select this mode (if supported) when Advanced Congestion Avoidance mode is inappropriate or not supported, or if fixed-traffic ratio apportionment or preferred/overflow behavior is needed.
Advanced Congestion Avoidance	A congestion-avoiding selection is made considering both primary and secondary candidate egress ports, ignoring the mix setting value. Egress ports with the minimal remaining hop count should be specified as primary; any suitable egress ports that have higher remaining hop counts should be specified as secondary. Candidate selection is based on vendor-specific metrics, favoring less-congested egress ports in general, and especially avoiding secondary candidates that are already heavily scheduled with primary traffic, regardless of the target DPID. An example congestion metric might be backlog-based, but with different weightings for primary vs. secondary backlogs. Congestion metric values for primary backlogs should be higher than secondary backlogs when assessing the congestion level of a secondary candidate egress port. This discourages the use of secondary candidate ports that have a high primary backlog. In congestion metrics, messages that are queued or internally routed via the physical port number in a DRT or via dynamic routing modes other than Advanced Congestion Avoidance should be considered primary backlog. The FM may choose to select this mode (if supported) for routing egress ports that carry commingled minimal and non-minimal traffic.

PBR switches that implement RGTs shall support at least one of the three architected dynamic routing modes (those listed in [Table 7-85](#)) within each RGT.

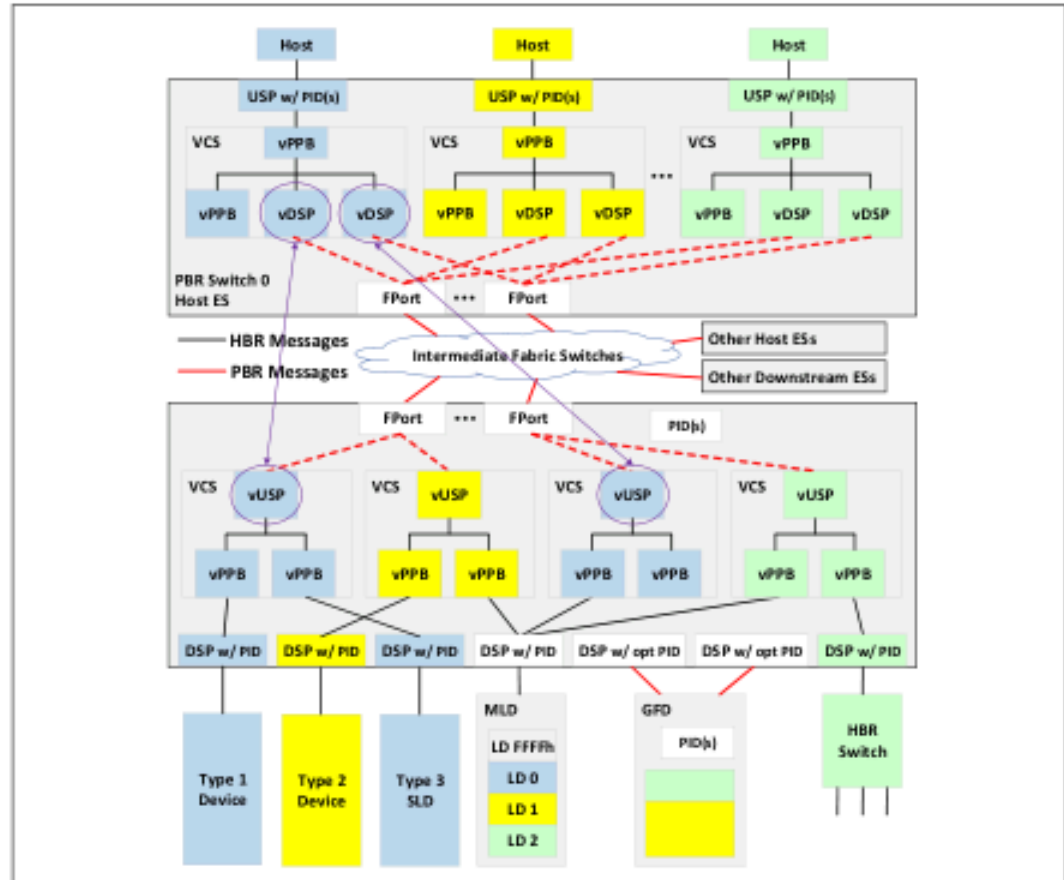
DRT entries that contain a single physical port instead of an RGT index are useful when there is only one reasonable egress port choice (e.g., routing to an Edge Port). This avoids an RGT look-up and additional processing to determine which egress port to use. This may also help reduce the number of entries that need to be implemented in the associated RGT.

7.7.6.4 PBR Switch vDSP/vUSP Bindings and Connectivity

Within the context of a single VH, the virtual connection between a VCS in the Host ES and a VCS in a Downstream ES is accomplished with a vDSP/vUSP binding. A vDSP is a vPPB in the Host ES VCS that the host sees as a DSP. A vUSP is a vPPB in the Downstream ES VCS that the host sees as a USP. Host software always sees a single virtual link connecting the vDSP and vUSP, even though one or more intermediate Fabric switches may be physically present.

Figure 7-46 shows an example PBR Fabric that consists of one Host ES, one Downstream ES, and an unspecified number of intermediate Fabric switches connecting the two.

Figure 7-46. Example PBR Fabric



The rules for vDSP/vUSP bindings are as follows:

- Each active Host ES vDSP is bound to one Host ES FPort and one Downstream ES vUSP
- Each active Downstream ES vUSP is bound to one Downstream ES FPort and one Host ES vDSP
- All messages routed using a vDSP/vUSP binding must contain both a DPID and an SPID
- vDSPs and vUSPs are never assigned PIDs
- Each PID used for vDSP/vUSP bindings may support both static and dynamic routing; however, vDSP/vUSP traffic always uses static routing
- Each vDSP/vUSP binding has a single host USP PID that determines which Host ES FPort will be used to route from vUSP to vDSP
- Each vDSP/vUSP binding has a single Downstream ES PID that determines which Downstream ES FPort will be used to route from vDSP to vUSP

When a Host ES FPort transmits a vDSP/vUSP message downstream in a PBR flit, the message contains the DPID and SPID taken from the vDSP's binding. Assuming no errors, the message traverses any intermediate Fabric switches that are present and is received by an FPort that is bound to the Downstream ES vUSP. A vUSP there claims the message by matching both the DPID and SPID from its binding.

Similarly, when a Downstream ES FPort transmits a vDSP/vUSP message upstream in a PBR flit, the message contains the DPID and SPID taken from the vUSP's binding. Assuming no errors, the message traverses any intermediate Fabric switches that are present and is received by an FPort that is bound to the Host ES vDSP. A vDSP there claims the message by matching both the DPID and SPID from its binding.

7.7.6.5 PID Use Models and Assignments

The example PBR Fabric illustrated in [Figure 7-46](#) illustrates key aspects of how PIDs can be assigned and used. PIDs are either assigned by the FM or by static fabric initialization (see [Section 7.7.12.1.1](#)).

A Host ES USP often has one PID but may have multiple PIDs assigned to support multiple vDSP/vUSP bindings in the same Downstream ES. Each vDSP/vUSP binding may use a different Host ES FPort and/or Downstream ES FPort, providing traffic isolation for differentiated quality of service. If multiple vDSP bindings use the same PID for the Downstream ES, different PIDs for the USP can distinguish their bindings.

The Downstream ES FPorts may have one or more PIDs assigned, where each PID can be associated with a different set of FPorts. In an example scenario, there might be one PID for the left set of FPorts for multipathing and another PID for the right set. For a PID assigned to an FPort set for multipathing, DRTs in different USPs can specify different egress ports for static routing, distributing the static routing traffic for certain topologies without requiring additional DS_ES PIDs.

A DSP may be assigned multiple PIDs, one PID, or no PIDs. A DSP above a non-GFD usually has one PID, but may be assigned multiple PIDs for isolating traffic from multiple senders or for associating a unique PID for each caching or HDM-DB-capable device attached to one or more HBR switches below an Edge Port. DSPs above a multi-ported GFD may not require dedicated assigned PIDs, relying instead on one or more PIDs assigned to the GFD itself.

A GFD may have one or more PIDs assigned. A multi-ported GFD may have multiple PIDs assigned for differentiated quality of service, though a single PID may be sufficient for congestion avoidance.

As mentioned in the previous section, each vDSP/vUSP binding has two PIDs assigned. For downstream vDSP/vUSP messages that use a given binding, the SPID is a PID associated with the host Edge USP, and the DPID is a PID associated with the Downstream ES FPort. Such messages are always transmitted by the same Host ES FPort and received by the same Downstream ES FPort. Then, the FPort uses various vUSP info decoding mechanisms to route the message to the appropriate Downstream ES vPPB using PBR mechanisms or HBR mechanisms, depending upon the message class. See CXL Switch Message Conversion (see [Section 7.7.6.6](#)). If there are any intermediate Fabric switches, such messages always take a single static path.

Upstream vDSP/vUSP messages are handled in a similar manner, but only involve CXL.io message classes. On a given binding, the SPID is the PID associated with the Downstream ES FPort, and the DPID is the PID associated with the host Edge USP. Such messages are always transmitted by the same Downstream ES FPort and received by the same Host ES FPort. Then, the receiving FPort uses the associated vDSP context to identify the appropriate target using HBR mechanisms. If the target is an egress port, the message is routed there for transmission. If the target is another vDSP, that vDSP

converts the PIDs to its bound PIDs and transmits it from its associated FPort, which may be the same FPort on which it arrived or on a different FPort. If there are any intermediate Fabric switches, such messages always take a single static path.

A PBR switch requires an assigned PID to send and receive management requests, responses, and notifications. Transactions that target this PID are processed by central logic or by FW within the switch.

FMs connected to a PBR switch via an MCTP-based CCI also consume a PID. This PID is communicated to the PBR switch when the FM claims ownership of the device. The PID is used to direct transactions to the FM, such as Event Notifications generated by components owned by the FM.

PID FFFh is reserved and is used to indicate that a transaction should be processed locally. It allows FMs to target devices before they have had a valid PID assigned and when they have an assigned PID of which the FM is unaware.

7.7.6.6 CXL Switch Message Format Conversion

A PBR switch converts messages received from HBR hosts, devices, and switches to the PBR message format for routing across a PBR Fabric. In addition, messages received from the PBR fabric that target the HBR hosts, devices, and switches are converted to messages using the non-PID spaces (i.e., CacheID, BI-ID, and LD-ID). The following subsections provide the conversion flow for each message class.

The FM assigns PIDs to various PBR switch ports, as described in [Section 7.7.6.5](#). The DPID value for request messages is determined by a variety of ways, including HDM Decoders, vPPB bindings, and lookup tables or CAMs using non-PID spaces. The DPID value for a response message is often the SPID value from the associated request message but is sometimes determined by one of the ways mentioned for request messages.

With HBR format messages, MLDs support a 4-bit LD-ID field in CXL.mem protocol for selection and routing of MLD messages, and CXL.cache includes a 4-bit CacheID field that is used to allow up to 16 Type 1 Devices or Type 2 Devices below an RP. PBR format messages use 12-bit PIDs to support large Fabrics. This section describes the support required in PBR switches for routing messages from non-fabric-aware hosts and devices that support the 4-bit LD-ID and 4-bit CacheID fields. It also covers BI-ID-based routing.

Considering the wide range of supported PBR/HBR switch topologies, the variety of specific routing techniques for the many different cases of port connectivity is quite complex. Below is a general description for the HBR and PBR switch routing mechanisms that are used by key message classes, followed by port processing tables with more-specific details for both classes of switches.

7.7.6.6.1 CXL.io, Including UIO

An HBR switch routes most CXL.io TLPs between its ports using standard mechanisms defined by PCIe Base Specification. A DSP above an MLD uses LD-ID Prefixes to identify which LD a downstream TLP is targeting or from which LD an upstream TLP came.

UIO Requests that directly target HDM ranges can use enhanced UIO-capable HDM Decoders for their routing. This includes UIO Requests from the host that target devices with HDM, as well as "Direct P2P" cases where UIO Requests from one device target other devices with HDM. UIO Direct P2P to HDM traffic goes upstream, P2P, and downstream along different portions of its path.

A PBR switch converts PCIe-format TLPs or CXL.io HBR-format TLPs to PBR-format TLPs by pre-pending to each TLP a 4B CXL PBR TLP Header (PTH), which includes an SPID and DPID. Conversion from PBR format to HBR format or PCIe format consists of stripping the CXL PTH from the TLP.

7.7.6.6.2 CXL.cache

A number of CXL.cache messages in 256B HBR format have a 4-bit CacheID field that enables up to 16 caching devices below a single RP. CXL.cache messages in 68B HBR format do not support this feature, and thus never carry a CacheID field. CXL.cache messages in PBR format do support this feature, but convey the necessary information via PIDs instead of a CacheID field. Table 7-86 summarizes which message classes contain the CacheID field.

Table 7-86. Summary of CacheID Field

Msg Class	CacheID Field		
	68B HBR	256B HBR	256B PBR
D2H Req	No	Yes	No
H2D Rsp	No	Yes	No
H2D DH	No	Yes	No
H2D Req	No	Yes	No
D2H Rsp	No	No	No
D2H DH	No	No	No

For HBR format messages that contain a CacheID field, in some cases an HBR or PBR DSP needs to know whether to propagate or assign the CacheID. This information is configured by host software and is contained in the CXL Cache ID Decoder Capability Structure (see Section 8.2.4.29).

Table 7-87 summarizes the HBR switch routing for CXL.cache message classes. Table 7-88 summarizes the PBR switch routing for CXL.cache message classes.

Table 7-87. Summary of HBR Switch Routing for CXL.cache Message Classes

Message Class	Switch Routing
D2H Request	For HBR switch routing of D2H requests upstream to the bound host, the D2H request to the USP relies on the DSP's vPPB binding at each switch level. CacheID is added to the message by the DSP above the device to enable routing of the H2D response.
H2D Response or Data Header	For HBR switch routing of H2D responses or data headers downstream to the DSP, the USP at each switch level looks up the PCIe-defined PortID from the Cache ID Route Table.
H2D Request	For HBR switch routing of H2D requests downstream to the DSP, the USP at each switch level looks up the PCIe-defined PortID from the Cache ID Route Table.
D2H Response or Data Header	For HBR switch routing of D2H responses or data headers upstream to the bound host, the D2H response or data header to the USP relies upon the DSP's vPPB binding at each switch level.

Within a PBR fabric, all CXL.cache messages are routed edge-to-edge, and they never use vDSP/vUSP bindings.

In contrast to most 256B HBR-format CXL.cache messages, PBR-format cache messages never contain a CacheID field, thus the equivalent information when needed must be conveyed via PIDs.

When multiple caching devices are attached to an HBR switch below a PBR fabric, the FM must allocate and assign a unique PID for each such caching device. This enables PBR switches to convert between a caching device's unique PID and CacheID when needed.

Table 7-88. Summary of PBR Switch Routing for CXL.cache Message Classes

Message Class	Switch Routing
D2H Request	For PBR switch routing of these messages upstream to the host, Edge DSPs get the Host USP DPID from their vPPB. Those above an SLD get their SPID from their vPPB. Those above an HBR USP look up the SPID from the Cache ID Route Table using the CacheID contained in the HBR-format message. For converting to HBR format at the Edge USP, the USP derives the CacheID from a 16-entry CAM using the SPID.
H2D Response or Data Header	For PBR switch routing of these messages downstream to the Edge DSP, the Edge USP looks up the DPID from the Cache ID Route Table using the CacheID in the HBR-format message. For converting to HBR format at the Edge DSP, above an SLD the CacheID is unused, and above an HBR USP the Cache ID is derived from a 16-entry CAM match using the DPID.
H2D Request	For PBR switch routing of these messages downstream to the Edge DSP, the Edge USP looks up the DPID from the CacheID Route Table using the CacheID. The USP gets the SPID from its vPPB. For converting to HBR format at the Edge DSP, above an SLD the CacheID is unused, and above an HBR USP the CacheID is derived from a 16-entry CAM match using the DPID.
D2H Response or Data Header	For PBR switch routing of these messages upstream to the host, Edge DSPs get the DPID from their vPPB. For converting to HBR format at the Edge USP, the CacheID field is not present in the message.

At an Edge DSP, when converting a downstream CXL.cache message from PBR to HBR format, if the CacheID field is unused, its value shall be cleared to 0.

7.7.6.6.3 CXL.mem

Several CXL.mem message classes in HBR format have a 4-bit LD-ID field that is used by Type 3 MLDs for determining the targeted LD. This feature is supported by both 68B and 256B HBR formats. PBR format conveys the necessary information via PIDs instead of an LD-ID field. Table 7-89 summarizes which message classes contain the LD-ID field.

Table 7-89. Summary of LD-ID Field

Msg Class	LD-ID Field		
	68B HBR	256B HBR	256B PBR
M2S Req	Yes	Yes	No
M2S RwD	Yes	Yes	No
S2M NDR	Yes	Yes	No
S2M DRS	Yes	Yes	No
S2M BISnp	N/A	In BI-ID	No
M2S BIRsp	N/A	In BI-ID	No

CXL.mem BISnp/BIRsp messages support the Back-Invalidate feature in 256B HBR format via a 12-bit BI-ID field, which determines the routing for BIRsp. This feature and its associated field are not supported in 68B HBR format. PBR format supports this feature and conveys the necessary information via 12-bit PIDs. Table 7-90 summarizes which message classes contain the BI-ID field.

In 256B HBR format over an MLD link, the 12-bit BI-ID field in BISnp/BIRsp carries the 4-bit LD-ID value, and the remaining 8 bits are all 0s. In 256B HBR format over non-MLD links, the 12-bit BI-ID field carries the 8-bit Bus Number of the HDM-DB device, and the remaining 4 bits are all 0s.

Table 7-90. Summary of BI-ID Field

Msg Class	BI-ID Field		
	68B HBR	256B HBR	256B PBR
S2M BISnp	N/A	Yes	No
M2S BIRsp	N/A	Yes	No

For messages that contain a BI-ID field, in some cases an HBR or PBR DSP needs to know whether to propagate or assign the BI-ID. This information is configured by host software and is contained in the CXL BI Decoder Capability Structure (see [Section 8.2.4.27](#)).

The Direct P2P CXL.mem for Accelerators use case, supported only by PBR fabrics, is not covered in this section; see [Section 7.7.10](#).

[Table 7-91](#) summarizes the HBR switch routing for CXL.mem message classes. [Table 7-92](#) summarizes the PBR switch routing for CXL.mem message classes.

Table 7-91. Summary of HBR Switch Routing for CXL.mem Message Classes

Message Class	Switch Routing
M2S Request	For HBR switch routing of M2S requests downstream toward the device, the HDM Decoder at the USP determines the PCIe-defined PortID of the DSP at each switch level. For a DSP above an MLD, there is a vPPB for each LD, which provides the LD-ID to insert in the request message.
S2M Response	For HBR switch routing of S2M responses upstream to the USP, the DSP relies on its vPPB binding at each switch level. For a DSP immediately above an MLD, there is a vPPB for each LD, and the LD-ID in the response message identifies the associated vPPB.
S2M BISnp	For HBR switch routing of S2M BISnp requests upstream to the USP, the DSP relies on its vPPB binding at each switch level. For a DSP immediately above an MLD, there is a vPPB for each LD, and the BI-ID in the response message carries an LD-ID that identifies the associated vPPB. The DSP then looks up the BusNum associated with its vPPB, places the BusNum in the BI-ID for later use in routing the associated BIRsp back to the DSP.
M2S BIRsp	For HBR switch routing of M2S BIRsp messages downstream to the DSP immediately above the device, the USP at each switch level relies on the BI-ID that carries the BusNum of the target DSP. The HBR switch then uses BusNum routing.

In an HBR switch, when filling in a subset of the bits in the BI-ID field with a value, the remaining bits in the BI-ID field shall be cleared to 0.

Within a PBR fabric, most CXL.mem message classes are routed edge-to-edge and do not use vDSP/vUSP bindings. The exceptions are M2S Req/RwD message classes with LD-FAM when host software has configured HDM Decoders in the Host ES USP to route them, in which case vDSP/vUSP bindings are used. See details regarding PBR Message Routing across the Fabric in [Section 7.7.6.2](#).

When HDM-DB devices are attached to an HBR switch below a PBR fabric, the FM must allocate and assign a unique PID for each HDM-DB device. This enables PBR switches to convert between an HDM-DB device's unique PID and Bus Number when needed.

Table 7-92. Summary of PBR Switch Routing for CXL.mem Message Classes

Message Class	Switch Routing
M2S Request	FAST/LDST Decoder Case: For Host ES routing of M2S requests downstream to the Edge DSP, the FAST/LDST decoder at the USP determines the DPID for routing the message edge-to-edge. HDM Decoder Case: For hierarchical routing of M2S requests downstream toward the Edge DSP, the HDM Decoder at the USP of each ES determines the egress vPPB (EvPPB), which contains an appropriate DPID. A vDSP in the Host ES contains the DPID/SPID that is used for targeting its partner Downstream ES vUSP. A DSP vPPB contains its dedicated DPID. Both host and Downstream ESs use PBR routing locally because a DSP above an MLD relies on the request having a valid SPID. For a DSP immediately above an MLD, a 16-entry CAM match using the SPID returns the associated LD-ID, which determines the LD-specific vPPB to use and is also inserted in the request message. For a DSP above a GFD, the message remains in PBR format.
S2M Response	For Edge DSP routing of S2M responses upstream to the Edge USP, the Edge DSP's vPPB contains the DPID for routing the message edge-to-edge. For a DSP immediately above an MLD, there is a vPPB for each LD, and the LD-ID in the response message identifies the associated vPPB. For a DSP above a GFD, the message is already in PBR format and remains so.
S2M BISnp	For Edge DSP routing of S2M BISnp messages upstream to the Edge USP, the Edge DSP's vPPB contains the DPID for routing the message edge-to-edge. For an Edge DSP immediately above an MLD, there is a vPPB for each LD, and the BI-ID in the BISnp carries an LD-ID that identifies the associated vPPB. The Edge DSP uses its vPPB's PID for the SPID. For an Edge DSP above an HBR USP, the BI-ID contains the BusNum associated with the HDM-DB device. The Edge DSP uses the BusNum to look up the associated SPID from a 256-entry table. At the Edge USP, the USP converts the BISnp to HBR format, copying the SPID value into the BI-ID.
M2S BIRsp	For Edge USP routing of M2S BIRsp messages downstream to the Edge DSP above the HDM-DB device, the Edge USP converts the BIRsp to PBR format, using the BI-ID value as the DPID, and then routes the BIRsp edge-to-edge. For an Edge DSP immediately above an MLD, a 16-entry CAM match using the SPID returns the associated LD-ID, which determines the LD-specific vPPB to use and is also inserted in the BI-ID field of the BIRsp. For an Edge DSP above an HBR switch USP, the DSP converts the BIRsp to HBR format, looking up the target BusNum in a 4k-entry table using the DPID, then copying it to the BI-ID. For a DSP above a GFD, the message remains in PBR format.

At an Edge DSP, when converting a downstream CXL.mem message from PBR to HBR format, if an LD-ID or BI-ID field is unused, its value shall be cleared to 0. Also, when filling in a subset of the bits in the BI-ID field with a value, the remaining bits in the BI-ID field shall be cleared to 0.

7.7.6.7 HBR Switch Port Processing of CXL Messages

Table 7-93, Table 7-94, and Table 7-95 summarize how HBR switches perform port processing of CXL.io, CXL.cache, and CXL.mem messages, respectively. A USP is below either an RP, a PBR DSP, or an HBR DSP. A USP can be in only one Virtual Hierarchy. A DSP is above either an HBR switch USP, an SLD, or an MLD.

For conciseness, there are many abbreviations within the tables. US stands for upstream. DS stands for downstream. P2P stands for peer-to-peer. DMA stands for direct memory access. Direct P2P stands for UIO Direct P2P to HDM. BusNum stands for Bus Number. "←" stands for assignment (e.g., "LD-ID Prefix ← vPPB context" means "the LD-ID prefix is assigned a value from the associated vPPB context"). Text beginning with "PCIe" (also shown in gold) means that the routing is defined in PCIe Base Specification.

In the CXL.io table (see Table 7-93), not all TLP types are explicitly covered; however, those not listed are usually handled by standard PCIe routing mechanisms (e.g., PCIe Messages are not explicitly covered, but ID-routed Messages are handled by PCIe ID routing, and address-routed Messages are handled by PCIe Memory Address routing).

Table 7-93. HBR Switch Port Processing Table for CXL.io

Message Class and Direction	HBR USP below RP or PBR/HBR DSP	HBR DSP		
		Above HBR USP	Above SLD	Above MLD
Cfg Req DS	PCIe ID routing	PCIe ID routing		PCIe ID routing LD-ID Prefix $\leftarrow$ vPPB context
Mem Req DS/US/P2P Incl UIO DMA Excl HDM UIO	PCIe Mem addr routing	PCIe Mem addr routing		PCIe Mem addr routing US: LD-ID Prefix identifies vPPB DS: LD-ID Prefix $\leftarrow$ vPPB context
HDM UIO Req Direct P2P and Host Requester	US: PCIe Mem addr routing DS: HDM Decoder routing	US: PCIe Mem addr routing DS/Direct P2P: USP HDM Decoder		US: PCIe Mem addr routing DS/Direct P2P: USP HDM Decoder US: LD-ID Prefix identifies vPPB DS: LD-ID Prefix $\leftarrow$ vPPB context
Cpl US	PCIe ID routing	PCIe ID routing		LD-ID Prefix identifies vPPB PCIe ID routing
Cpl DS	PCIe ID routing	PCIe ID routing		PCIe ID routing LD-ID Prefix $\leftarrow$ vPPB context

Table 7-94. HBR Switch Port Processing Table for CXL.cache

Message Class and Direction	HBR USP below RP or PBR/HBR DSP	HBR DSP		
		Above HBR USP	Above SLD	Above MLD
D2H Req US	Propagate CacheID	Propagate CacheID vPPB binding routing to USP	CacheID←Local Cache ID field vPPB binding routing to USP	
H2D Rsp/DH DS	Propagate CacheID PortID←Cache ID Route Table	Propagate CacheID	Propagate Cache ID (SLD should ignore it)	
H2D Req DS	PortID routing to DSP OS must handle multi-level HBR			
D2H Rsp/DH US	-	vPPB binding routing to USP		

Table 7-95. HBR Switch Port Processing Table for CXL.mem

Message Class and Direction	HBR USP below RP or PBR/HBR DSP	HBR DSP		
		Above HBR USP	Above SLD	Above MLD
M2S Req DS	PortID ← HDM Decoder (HPA) Routing to DSP uses PortID	Propagate LD-ID (not used by these receivers)		LD-ID ← vPPB context
S2M Rsp US	Propagate LD-ID (not used by these receivers)	vPPB binding routing to USP Propagate LD-ID (not used for internal switch routing)		LD-ID identifies vPPB vPPB binding routing to USP
S2M BISnp US	BI-ID[7:0] contains BusNum Propagate BI-ID	Propagate BI-ID vPPB binding routing to USP	Received BI-ID is ignored BI-ID[7:0] ← BusNum(vPPB) vPPB binding routing to USP	BI-ID[3:0] contains LD-ID LD-ID identifies vPPB BI-ID[7:0] ← BusNum(vPPB) vPPB binding routing to USP
M2S BIRsp DS	Target BusNum ← BI-ID[7:0] PCIe BusNum routing to DSP	Propagate BI-ID	Propagate BI-ID (SLD should ignore it)	BI-ID[3:0] ← LD-ID(vPPB)

7.7.6.8 PBR Switch Port Processing of CXL Messages

Table 7-96, Table 7-97, and Table 7-98 summarize how PBR switches perform port processing of CXL.io, CXL.cache, and CXL.mem messages, respectively. A PBR USP must be below an RP and can be in only one Virtual Hierarchy. A PBR DSP is above either an SLD, an MLD, a GFD, or an HBR switch USP. A PBR FPort can only be connected to another PBR FPort in a different PBR switch.

For conciseness, there are many abbreviations within the tables. US stands for upstream. DS stands for downstream. P2P stands for peer-to-peer. DMA stands for direct memory access. Direct P2P stands for UIO Direct P2P to HDM. EvPPB stands for Egress vPPB. BusNum stands for Bus Number. RT stands for the CacheID Route Table. "←" stands for assignment (e.g., "LD-ID Prefix ← vPPB context" means "the LD-ID prefix is assigned a value from the associated vPPB context"). Also referring to a vPPB context, vPPB.root.PID stands for the PID of the associated Edge USP, and vPPB.self.PID stands for the PID of the vPPB itself. Eg2Eg means Edge-to-Edge. Text beginning with "PCIe" (also shown in gold) means that the routing is defined in PCIe Base Specification.

In the CXL.io table (see Table 7-96), not all TLP types are explicitly covered; however, those not listed are usually handled by standard PCIe routing mechanisms (e.g., PCIe Messages are not explicitly covered, but ID-routed Messages are handled by PCIe ID routing, and address-routed Messages are handled by PCIe Memory Address routing).

In the CXL.mem table (see Table 7-98) the Direct P2P CXL.mem for Accelerators use case is not covered; see Section 7.7.10.3.

Table 7-96. PBR Switch Port Processing Table for CXL.io (Sheet 1 of 2)

Message Class and Direction	Edge USP Always below an RP	Host ES FPort with vDSP(s)	Downstream ES FPort with vUSP(s)	Edge DSP in Either Host ES or Downstream ES			
				Above HBR Switch USP	Above SLD	Above MLD	Above GFD
Cfg Req DS	PCIe ID routing to DSP or vDSP	vDSP converts to PBR fmt; FPort xmits to vUSP's FPort	vUSP matches DPID/SPID; vUSP converts to HBR fmt; PCIe ID routing to DSP	PCIe ID routing		PCIe ID routing LD-ID Prefix ← vPPB LD-ID	N/A
Mem Req DS/US/P2P Incl UIO DMA Excl HDM UIO	PCIe Mem addr routing	DS: vDSP converts to PBR fmt; FPort xmits to vUSP's FPort US/P2P: vDSP matches DPID/SPID; vDSP converts to HBR fmt; PCIe Mem addr routing	DS: vUSP matches DPID/SPID; vUSP converts to HBR fmt; PCIe Mem addr routing US: vUSP converts to PBR fmt; FPort xmits to vDSP's FPort	PCIe Mem addr routing		PCIe Mem addr routing DS: LD-ID Prefix ← vPPB.LD-ID US/P2P: LD-ID Prefix identifies vPPB	N/A
Cpl US/P2P Excl HDM UIO	PCIe ID routing	vDSP matches DPID/SPID; vDSP converts to HBR fmt; PCIe ID routing	vUSP converts to PBR fmt; FPort xmits to vDSP's FPort	PCIe ID routing		LD-ID Prefix identifies vPPB PCIe ID routing	N/A
Cpl DS Excl HDM UIO	PCIe ID routing to DSP or vDSP	vDSP converts to PBR fmt; FPort xmits to vUSP's FPort	vUSP matches DPID/SPID; vUSP converts to HBR fmt; PCIe ID routing to DSP	PCIe ID routing		PCIe ID routing LD-ID Prefix ← vPPB.LD-ID	N/A
HDM UIO Req HDM Decoder case for Direct P2P and Host Requester	Direct P2P: N/A Host Requester (DS): HDM Decoder routes to DSP or vDSP	DS: vDSP converts to PBR fmt; FPort xmits to vUSP's FPort US/P2P: vDSP matches DPID/SPID; vDSP converts to HBR fmt; USP's HDM Decoder routes P2P	DS: vUSP matches DPID/SPID; vUSP converts to HBR fmt; HDM Decoder routes to DSP US: vUSP converts to PBR fmt; FPort xmits to vDSP's FPort	US/P2P: If above MLD, LD-ID Prefix identifies vPPB; USP/vUSP HDM Decoder routes US or P2P within same switch DS: Convert to HBR fmt; if above MLD, LD-ID Prefix ← vPPB.LD-ID			N/A

Table 7-96. PBR Switch Port Processing Table for CXL.io (Sheet 2 of 2)

Message Class and Direction	Edge USP Always below an RP	Host ES FPort with vDSP(s)	Downstream ES FPort with vUSP(s)	Edge DSP in Either Host ES or Downstream ES			
				Above HBR Switch USP	Above SLD	Above MLD	Above GFD
HDM UIO Cpl HDM Decoder case for Direct P2P and Host Requester	Direct P2P: N/A Host Requester (US): PCIe ID routing	vDSP matches DPID/ SPID; vDSP converts to HBR fmt; PCIe ID routing	vUSP converts to PBR fmt; FPort xmits to vDSP's FPort	US: If above MLD, LD-ID Prefix identifies vPPB; PCIe ID routing DS: PCIe ID routing ; if above MLD, LD-ID Prefix $\leftarrow$ vPPB.LD-ID			N/A
HDM UIO Req FAST/LDST case for Direct P2P and Host Requester	Direct P2P: N/A Host Requester (DS): FAST/LDST converts to PBR and routes Eg2Eg	Route Eg2Eg	Route Eg2Eg	US/P2P: If above MLD, LD-ID Prefix identifies vPPB; FAST/LDST converts to PBR and routes Eg2Eg DS: Convert to HBR fmt; if above MLD, LD-ID Prefix $\leftarrow$ CAM ₁₆ (SPID)			US: N/A DS: Keep in PBR
HDM UIO Cpl FAST/LDST case for Direct P2P and Host Requester	Direct P2P: N/A Host Requester (US): Convert to HBR	Route Eg2Eg	Route Eg2Eg	US: If above MLD, LD-ID Prefix identifies vPPB; convert to PBR using UIO ID-based Rerouter; route Eg2Eg DS: Convert to HBR			US: Keep in PBR; route Eg2Eg DS: N/A

Table 7-97. PBR Switch Port Processing Table for CXL.cache

Message Class and Direction	Edge USP Always below an RP	Host ES FPort with vDSP(s)	Downstream ES FPort with vUSP(s)	Edge DSP in Either Host ES or Downstream ES			
				Above HBR Switch USP	Above SLD	Above MLD	Above GFD
D2H Req US	Convert to HBR fmt CacheID $\leftarrow$ CAM ₁₆ (SPID)	Route Eg2Eg	Route Eg2Eg	Convert to PBR fmt DPID $\leftarrow$ vPPB.root.PID SPID $\leftarrow$ RT(CacheID)	Convert to PBR fmt DPID $\leftarrow$ vPPB.root.PID SPID $\leftarrow$ vPPB.self.PID		
H2D Rsp/DH DS	Convert to PBR fmt DPID $\leftarrow$ RT(CacheID)	Route Eg2Eg	Route Eg2Eg	Convert to HBR fmt 256B: CacheID $\leftarrow$ CAM ₁₆ (DPID) 68B: Has no CacheID	Convert to HBR fmt 256B: CacheID $\leftarrow$ 0 68B: Has no CacheID		
H2D Req DS	Convert to PBR fmt DPID $\leftarrow$ RT(CacheID) SPID $\leftarrow$ vPPB.self.PID	Route Eg2Eg	Route Eg2Eg				
D2H Rsp/DH US	Convert to HBR fmt	Route Eg2Eg	Route Eg2Eg	Convert to PBR fmt DPID $\leftarrow$ vPPB.root.PID	Convert to PBR fmt DPID $\leftarrow$ vPPB.root.PID		

Table 7-98. PBR Switch Port Processing Table for CXL.mem

Message Class and Direction	Edge USP Always below an RP	Host ES FPort with vDSP(s)	Downstream ES FPort with vUSP(s)	Edge DSP in Either Host ES or Downstream ES			
				Above HBR Switch USP	Above SLD	Above MLD	Above GFD
M2S Req/Resp DS	FAST or LDST: Convert to PBR fmt DPID ← xxST(HPA) SPID ← vPPB.self.PID	Route Eg2Eg	Route Eg2Eg	Convert to HBR fmt LD-ID ← 0; is unused		LD-ID ← CAM ₁₆ (SPID) Convert to HBR MLD fmt	LD-ID is N/A Keep in PBR fmt
	HDM Decoder: Convert to PBR fmt EvPPB ← HDM-Dec(HPA) DPID ← EvPPB.bndg.PID SPID ← vPPB.self.PID Route to local DSP or vDSP FPort	vDSP's FPort xmits to vUSP's FPort	vUSP matches DPID/SPID vUSP keeps in PBR fmt EvPPB ← HDM-Dec(HPA) DPID ← EvPPB.self.PID Route to egress DSP				N/A
S2M NDR/DRS US	Convert to HBR fmt LD-ID ← 0; is unused	Route Eg2Eg	Route Eg2Eg	LD-ID is unused Convert to PBR fmt DPID ← vPPB.root.PID		LD-ID identifies vPPB Convert to PBR fmt DPID ← vPPB.root.PID	Keep in PBR fmt LD-ID is N/A
S2M BISnp US	Convert to HBR fmt BI-ID[11:0] ← SPID	Route Eg2Eg	Route Eg2Eg	Convert to PBR fmt DPID ← vPPB.root.PID BusNum ← BI-ID[7:0] SPID ← RAM ₂₅₆ (BusNum)	Convert to PBR fmt DPID ← vPPB.root.PID SPID ← vPPB.self.PID	BI-ID[3:0] contains LD-ID LD-ID identifies vPPB Convert to PBR fmt DPID ← vPPB.root.PID SPID ← vPPB.self.PID	Keep in PBR fmt
M2S BIRsp DS	Convert to PBR fmt DPID ← BI-ID[11:0]	Route Eg2Eg	Route Eg2Eg	Convert to HBR fmt BusNum ← RAM _{4k} (DPID) BI-ID[7:0] ← BusNum	Convert to HBR fmt BI-ID is unused	Convert to HBR fmt LD-ID ← CAM ₁₆ (SPID) BI-ID[3:0] ← vPPB.LD-ID	Keep in PBR fmt

7.7.6.9 PPB and vPPB Behavior of PBR Link Ports

A PBR Link port has two varieties: an Inter-Switch Link (ISL) and a GFD Link.

The ISL case is a downstream-to-downstream crosslink. The DSP on each side of the link is managed by the FM with assistance from switch firmware. The full PCIe capabilities of a DSP shall be available. Bus master enable, AER, DPC, and other capabilities that an host typically controls will be controlled by the FM and/or switch firmware.

Other users of an ISL can be any number of VHS. The ISL (and as many switch hops and additional ISLs as it takes) functions as a single link between vDSP and vUSP. Any one ISL can potentially be shared by multiple VHS. Because a VH shares the link with other VHS, there is no way for a VH to control any of the link physical characteristics. However, the Host ES vDSP shall reflect the physical link settings for the fabric port DSP to which it is bound (e.g., link speed, lane margining, etc.).

A GFD PBR link is similar to an ISL in that many VH can share it. It is different however in that no vDSP nor vUSP is associated with it. The link itself is a simple up-down link, with the switch having an (FM-owned) DSP leading, via the PBR link, to the USP of a GFD. A switch DSP should have full PCIe capabilities, just like for an ISL or any other DSP.

The remainder of this section focuses on the vDSP and vUSP perspective, from the PCIe configuration space, for a variety of capabilities:

- "Supported" means that the PCIe register is available to be read and written by the host
- "Not supported" means that the register is either read-only or the capability is unavailable
- "Mirrors DSP" means that the values reflect the (typically physical link) value in the DSP
- "Read/Write with no effect" implies that the vDSP/vUSP register will be unaffected by reads and writes

It is expected that a fabric port DSP supports all PCIe capabilities required by the PCIe spec for a downstream port. DPC, which is optional for PCIe, is required for CXL for a DSP that is a fabric port.

7.7.6.9.1 ISL Type 1 Configuration Space Header

Table 7-99. ISL Type 1 Configuration Space Header

Register	Register Fields	FM-owned DSP	vDSP	vUSP
Bridge Control Register	Parity Error Response Enable	Supported	Supported	Supported
	SERR# Enable	Supported	Supported	Supported
	ISA Enable	Not Supported	Not Supported	Not Supported
	Secondary Bus Reset	Supported	Supported	Supported

7.7.6.9.2 ISL PCIe-compatible Configuration Register

Table 7-100. ISL PCIe Configuration Space Header

Register	Register Fields	FM-owned DSP	vDSP	vUSP
Command	I/O Space Enable	Hardwire to 0	Supported	Supported
Command	Memory Space Enable	Supported	Supported	Supported
	Bus Master Enable	Not Supported	Supported	Supported
	Parity Error Response	Supported	Supported	Supported
	SERR# Enable	Supported	Supported	Supported
	Interrupt Disable	Supported	Supported	Supported
Status	Interrupt Status	Hardwire to 0	Supported	Supported
	Master Data Parity Error	Supported	Supported	Supported
	Signaled System Error	Supported	Supported	Supported
	Detected Parity Error	Supported	Supported	Supported

7.7.6.9.3 ISL PCIe Capability Structure

Table 7-101. ISL PCIe Capability Structure (Sheet 1 of 3)

Register	Register Fields	FM-owned DSP	vDSP	vUSP
Device Capabilities	Max Payload Size Supported	FM Configured	Mirrors DSP	Mirrors DSP
	Phantom Functions Supported	0	0	0
	Extended Tag Field Supported	Supported	Supported	Supported
Device Control	Max Payload Size	FM Configured	Mirrors DSP	Mirrors DSP
Link Capabilities	Link Bandwidth Notification Capability	0	0	0
	ASPM Support	No L0s	no L0s	no L0s
	Clock Power Management	No PM L1 Substates	Mirrors DSP	Mirrors DSP
Link Control	ASPM Control	Supported	Not Supported	Not Supported
	Link Disable	Supported	Supported	Not Supported
	Retrain Link	Supported	Read/Write with no effect	Not Supported
	Common Clock Configuration	Supported	Read/Write with no effect	
	Extended Synch	Supported	Read/Write with no effect	
	Hardware Autonomous Width Disable	Supported	Read/Write with no effect	
	Link Bandwidth Management Interrupt Enable	Supported	Read/Write with no effect	Not Supported
	Link Autonomous Bandwidth Interrupt Enable	Supported	Supported	Not Supported
	Flit Mode Disable	0	0	0
	DRS Signaling Control	Supported	Supported	Not Supported

Table 7-101. ISL PCIe Capability Structure (Sheet 2 of 3)

Register	Register Fields	FM-owned DSP	vDSP	vUSP
Link Status	Current Link Speed	Supported	Mirrors DSP	Mirrors DSP
	Negotiated Link Speed	Supported	Mirrors DSP	Mirrors DSP
	Link Training	Supported	0	0
	Slot Clock Configuration	Supported	Mirrors DSP	Mirrors DSP
	Data Link Layer Active	Supported	Mirrors DSP	0
	Link Bandwidth Management Status	Supported	Mirrors DSP	0
	Link Autonomous Bandwidth Status	Supported	Mirrors DSP	0
Slot Capabilities	Hot-Plug Surprise	Supported	Mirrors DSP	0
	Physical Slot Number	Supported	Supported	0
Slot Status	Attention Button Pressed	Supported	Supported	0
	Power Fault Detected	Supported	Mirrors DSP	0
	MRL Sensor Changed	Supported	Mirrors DSP	0
	Presence Detect Changed	Supported	Supported	0
	MRL Sensor State	Supported	Mirrors DSP	0
	Presence Detect State	Supported	Supported	0
	Electromechanical Interlock Status	Supported	Mirrors DSP	0
	Data Link Layer State Changed	Supported	Supported	0
Device Capabilities 2	All bits	Supported	Supported	0
Device Control 2	ARI Forwarding Enable	Supported	Supported	0
	Atomic Op Egress Blocking	Supported	Supported	0
	LTR Mechanism Enabled	Supported	Supported	0
	Emergency Power Reduction Request	Supported	Read/Write with no effect	0
	End-End TLP Prefix Blocking	Supported	Mirrors DSP. Read/Write with no effect	0
Link Control 2	Target Link Speed	Supported	Read/Write with no effect	Read/Write with no effect
	Enter Compliance	Supported	Read/Write with no effect	Read/Write with no effect
	Hardware Autonomous Speed Disable	Supported	Read/Write with no effect	Read/Write with no effect
	Selectable De-emphasis	Supported	Read/Write with no effect	Read/Write with no effect
	Transmit Margin	Supported	Read/Write with no effect	Read/Write with no effect
	Enter Modified Compliance	Supported	Read/Write with no effect	Read/Write with no effect
	Compliance SOS	Supported	Read/Write with no effect	Read/Write with no effect
	Compliance Preset/De-emphasis	Supported	Read/Write with no effect	Read/Write with no effect

Table 7-101. ISL PCIe Capability Structure (Sheet 3 of 3)

Register	Register Fields	FM-owned DSP	vDSP	vUSP
Link Status 2	Current De-emphasis Level	Supported	Mirrors DSP	Mirrors DSP
	Equalization 8.0 GT/s Complete	Supported	Mirrors DSP	Mirrors DSP
	Equalization 8.0 GT/s Phase 1 Successful	Supported	Mirrors DSP	Mirrors DSP
	Equalization 8.0 GT/s Phase 2 Successful	Supported	Mirrors DSP	Mirrors DSP
	Equalization 8.0 GT/s Phase 3 Successful	Supported	Mirrors DSP	Mirrors DSP
	Link Equalization Request 8.0 GT/s	Supported	Read/Write with no effect	Read/Write with no effect
	Retimer Presence Detected	Supported	Mirrors DSP	Mirrors DSP
	Two Retimers Presence Detected	Supported	Mirrors DSP	Mirrors DSP
	Crosslink Resolution	Supported	All 0s	All 0s
	Flit Mode Status	Supported	Supported	Supported
	Downstream Component Presence	Supported	Supported	0
	DRS Message Received	Supported	Supported	0

7.7.6.9.4 ISL Secondary PCIe Capability Structure

All fields in the Secondary PCIe Capability Structure for a Virtual PPB shall behave identically to PCIe except the following:

Table 7-102. ISL Secondary PCIe Extended Capability

Register	Register Fields	FM-owned DSP	vDSP	vUSP
Link Control 3	Perform Equalization	Supported	Read/Write with no effect	Read/Write with no effect
	Link Equalization Request Interrupt Enable	Supported	Read/Write with no effect	Read/Write with no effect
	Enable Lower SKP OS Generation Vector	Supported	Read/Write with no effect	Read/Write with no effect
Lane Error Status	All fields	Supported	Mirrors DSP	Mirrors DSP
Lane Equalization Control	All fields	Supported	Read/Write with no effect	Read/Write with no effect
Data Link Features Capabilities	All fields	Supported	Mirror DSP	Mirror DSP

7.7.6.9.5 ISL Physical Layer 16.0 GT/s Extended Capability

All fields in the Physical Layer 16.0 GT/s Extended Capability Structure for a Virtual PPB shall behave identically to PCIe except the following:

Table 7-103. ISL Physical Layer 16.0 GT/s Extended Capability

Register	Register Fields	FM-owned DSP	vDSP	vUSP
16.0 GT/s Status	All fields	Supported	Mirrors DSP	Mirrors DSP
16.0 GT/s Local Data Parity Mismatch Status	Local Data Parity Mismatch Status	Supported	Mirrors DSP	Mirrors DSP
16.0 GT/s First Retimer Data Parity Mismatch Status	First Retimer Data Parity Mismatch Status	Supported	Mirrors DSP	Mirrors DSP
16.0 GT/s Second Retimer Data Parity Mismatch Status	Second Retimer Data Parity Mismatch Status	Supported	Mirrors DSP	Mirrors DSP
16.0 GT/s Lane Equalization Control	Downstream Port 16.0 GT/s Transmitter Preset	Supported	Mirrors DSP	Mirrors DSP

7.7.6.9.6 ISL Physical Layer 32.0 GT/s Extended Capability

All fields in the Physical Layer 32.0 GT/s Extended Capability Structure for a Virtual PPB shall behave identically to PCIe except the following:

Table 7-104. ISL Physical Layer 32.0 GT/s Extended Capability

Register	Register Fields	FM-owned DSP	vDSP	vUSP
32.0 GT/s Control Register	All fields	Supported	Read/Write with no effect	Read/Write with no effect
32.0 GT/s Status Register	Link Equalization Request 32.0 GT/s	Supported	Read/Write with no effect	Read/Write with no effect
	All fields except Link Equalization Request 32.0 GT/s	Supported	Mirrors DSP	Mirrors DSP
Received Modified TS Data 1 Register	All fields	Supported	Mirrors DSP	Mirrors DSP
Received Modified TS Data 2	All fields	Supported	Mirrors DSP	Mirrors DSP
Transmitted Modified TS Data 1	All fields	Supported	Mirrors DSP	Mirrors DSP
32.0 GT/s Lane Equalization Control	Downstream Port 32.0 GT/s Transmitter Preset	Supported	Mirrors DSP	Mirrors DSP

7.7.6.9.7 ISL Physical Layer 32.0 GT/s Extended Capability

All fields in the Physical Layer 64.0 GT/s Extended Capability Structure for a Virtual PPB shall behave identically to PCIe except the following:

Table 7-105. ISL Physical Layer 64.0 GT/s Extended Capability

Register	Register Fields	FM-owned DSP	vDSP	vUSP
64.0 GT/s Control Register	All fields	Supported	Read/Write with no effect	Read/Write with no effect
64.0 GT/s Status Register	Link Equalization Request 64.0 GT/s	Supported	Read/Write with no effect	Read/Write with no effect
	All fields except Link Equalization Request 64.0 GT/s	Supported	Mirrors DSP	Mirrors DSP
Received Modified TS Data 1 Register	All fields	Supported	Mirrors DSP	Mirrors DSP
Received Modified TS Data 2	All fields	Supported	Mirrors DSP	Mirrors DSP
Transmitted Modified TS Data 1	All fields	Supported	Mirrors DSP	Mirrors DSP
64.0 GT/s Lane Equalization Control	Downstream Port 64.0 GT/s Transmitter Preset	Supported	Mirrors DSP	Mirrors DSP

7.7.6.9.8 ISL Lane Margining at the Receiver Extended Capability

All fields in the ISL Lane Margining at the Receiver for a Virtual PPB shall behave identically to PCIe except the following:

Table 7-106. ISL Lane Margining at the Receiver Extended Capability

Register	Register Fields	FM-owned DSP	vDSP	vUSP
Margining Port Status Register	All fields	Supported	Mirrors DSP	Mirrors DSP
Margining Lane Control Register	All fields	Supported	Read/Write with no effect	Read/Write with no effect

7.7.6.9.9 ISL ACS Extended Capability

ACS applies only to a Downstream Port which, for a PBR link, applies to either a DSP above a GFD, a DSP connected to a crosslink, or a vDSP in a VH. All fields in the ISL ACS at the Receiver for a Virtual PPB shall behave identically to PCIe.

7.7.6.9.10 ISL Advanced Error Reporting Extended Capability

AER can apply to a vPPB on any side of a link. FM-owned DSPs, vDSPs, and vUSPs support all AER fields.

7.7.6.9.11 ISL DPC Extended Capability

DPC for both vDSP and vUSP is supported for all fields. The FM-owned DSP above an ISL must have DPC. DPC on the DSP above an ISL shall always be enabled by FM. DPC support is required to provide sufficient delay so that the various software entities — switch firmware, host software, fabric manager — are able to complete DPC event processing at their own pace.